NAND to Tetris-I

Lab 1: Digital Logic & ALU Design

Name: Aditya Agrawal
Roll No: 2501410078
Email: 2501410078@krmu.edu.in

Q1. Q-1: Implement NOT, AND, OR gates (5 marks)

module and_nand(input A, input B, output Y CHIP Not { IN in; Out out; Parts: Nand(a = in , b = in, out=out); } CHIP and { In a,b; Out out; Parts: Nand (a = a , b = b , out = nandout); } CHIP OR gates { In a,b; Out out; Parts: Nand(a=a,b=a, out =notA); Nand(a=b,b=b, out =notB); Nand(a=out,b=out, out=out); }

Q2. Q-2 (5 marks)

CHIP Xor { IN a,b; Out out; Parts: Or (a=a, b=b, out=orAB); And (a=a, b=b, out=andAb); Not (a=a, b=b, out=not And); And(a=or AB ,b=notAnd, out=out); } CHIP Mux { In a,b , sel: out out; parts: Not(in=set, out=notsel): And(a=a, b=not sel, out=aAnd); And(a=b, b=sel, out=bAnd); or(a=aAnd, b=bAnd, Out=out); } CHIPMux4-way16 { IN a[16], b[16], c[16], d[16], sel[2]; Out out[16]